

Plasmonic Surface Lattice Resonances: Theory and Computation

Published as part of the *Accounts of Chemical Research* special issue “*Nanochemistry for Plasmonics and Plasmonics for Nanochemistry*”.

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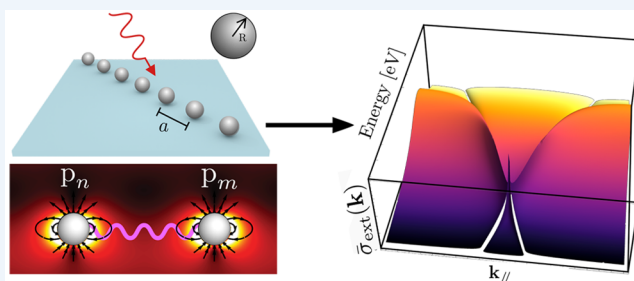
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CONSPECTUS: Plasmonic surface lattice resonances (SLRs) are mixed light–matter states emergent in a system of periodically arranged metallic nanoparticles (NPs) under the constraint that the array spacing is able to support a standing wave of optical-frequency light. The properties of SLRs derive from two separate physical effects; the electromagnetic (plasmonic) response of metal NPs and the electromagnetic states (photonic cavity modes) associated with the array of NPs.

Metal NPs, especially free-electron metals such as silver, gold, aluminum, and alkali metals, support optical-frequency electron density oscillations known as localized surface plasmons (LSPs). The high density of conduction-band electrons in these metals gives rise to plasmon excitations that strongly couple to light even for particles that are several orders of magnitude smaller than the wavelength of the excitation source. In this sense, LSPs have the remarkable ability to squeeze far-field light into intensely localized electric near-fields that can enhance the intensity of light by factors of $\sim 10^3$ or more. Moreover, as a result of advances in the synthesis and fabrication of NPs, the intrinsic dependence of LSPs on the NP geometry, composition, and size can readily be exploited to design NPs with a wide range of optical properties. One drawback in using LSPs to enhance optical, electronic, or chemical processes is the losses introduced into the system by dephasing and Ohmic damping—an effect that must either be tolerated or mitigated. Plasmonic SLRs enable the mitigation of loss effects through the coupling of LSPs to diffractive states that arise from arrays satisfying Bragg scattering conditions, also known as Rayleigh anomalies. Bragg modes are well-known for arrays of dielectric NPs, where they funnel and trap incoming light into the plane of the lattice, defining a photonic cavity. The low losses and narrow linewidths associated with dielectric NPs produce Bragg modes that oscillate for $\sim 10^3$ – 10^4 cycles before decaying. These modes are of great interest to the metamaterials community but have relatively weak electric fields associated with dielectric NPs and therefore are not used for applications where local field enhancements are needed.

Plasmonic lattices, i.e., photonic crystals composed of metallic NPs, combine the characteristics of both LSPs and diffractive states, enabling both enhanced local fields and narrow-linewidth excitations, in many respects providing the best advantages of both materials. Thus, by control of the periodicity and global symmetry of the lattice in addition to the material composition and shape of the constituent NPs, SLRs can be designed to simultaneously survive for up to 10^3 cycles while maintaining the electric field enhancements near the NP surface that have made the use of LSPs ubiquitous in nanoscience.

Modern fabrication methods allow for square-centimeter-scale patches of two-dimensional arrays that are composed of approximately one trillion NPs, making them effectively infinite at the nanoscale. Because of these advances, it is now possible to experimentally realize SLRs with properties that approach those predicted by idealized theoretical models. In this Account, we introduce the fundamental theory of both SLRs and SLR-mediated lasing, where the latter is one of the most important applications of plasmonic SLRs that has emerged to date. The focus of this Account is on theoretical concepts for describing plasmonic SLRs and computational methods used for their study, but throughout we emphasize physical insights provided by the theory that aid in making applications.



INTRODUCTION

Since the discovery of graphene,¹ the study of the optical and electronic properties of novel two-dimensional (2D) materials has been at the forefront of modern chemistry and materials science. In particular, 2D materials composed of group IV elements arranged into honeycomb lattices, of which graphene is the prototypical example, are well-known for unusual electronic

properties arising from degeneracies at high-symmetry points of the band structure.² Fabricating lattices with *nanoparticles* (NPs) in place of atoms and treating the multipolar electromagnetic response of the NPs similarly to the role of atomic wave

Received: June 14, 2019

Published: August 29, 2019

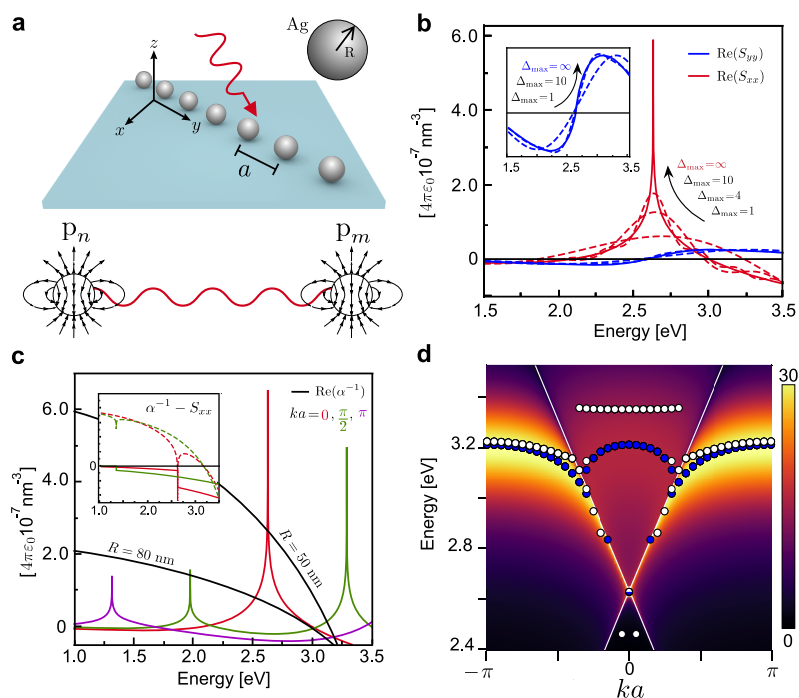


Figure 1. 1D chain array of 50 nm radius Ag NPs with periodicity $a = 470$ nm and parametrized using the Drude model by setting $\epsilon_{\infty} = 5.57$, $\gamma = 0.018$ eV, and $\omega_p = 9.615$ eV. (a) Schematic of the 1D chain and the interaction between transverse-oriented dipoles sharing a long-range photon. (b) Lattice sums for transversely and longitudinally polarized SLRs for varying values of the maximum number of nearest-neighbor interactions, Δ_{max} . (c) Lattice sums for transverse polarization and varying values of ka . The inset plots the real and imaginary parts of the complex-valued function $[\alpha^{-1} - S_{xx}]$. When $[\alpha^{-1} - S_{xx}] = 0$, an SLR mode exists. (d) TE-polarized extinction spectra with overlaid $\text{Re}[\hbar\omega_{\text{LP}}]$ for the cases where the imaginary part of the eigenequation is ignored (blue) or included (white). Both methods lead to a degenerate mode at $ka = 0$ (mixed white/blue).

functions yield *photonic* rather than electronic band structure.³ Nontrivial 2D photonic band structures have been achieved using both dielectric and metallic NPs. Lattices composed of metal NPs are of particular interest, as the particles support strong collective surface-bound oscillations of conduction-band electrons known as localized surface plasmons (LSPs) in the UV and visible spectral regions.⁴ The strong subdiffraction-limited electric fields produced with LSPs are used to enhance and control light–matter interactions with nearby light emitters and absorbers.⁵ Indeed, LSPs can enhance the local electric field seen by a nearby molecule by factors of 10^3 or more,⁶ making their use ubiquitous in diverse disciplines ranging from classical optics to quantum optics and spectroscopy as well as in chemical, storage, and biosensing applications.

When arranged into 2D periodic lattices, metallic NPs interact with one another through their LSP-induced electric fields, and when the distance between the NPs becomes approximately commensurate with the wavelength of optical light (divided by the local index), these collective LSP modes take on the diffractive character of the 2D photonic cavity modes supported by a lattice of nonresonant scatterers.^{7,8} The photonic normal modes of these systems are called surface lattice resonances (SLRs), so the term *plasmonic SLR* is appropriate for describing them. We will drop the adjective “plasmonic” in this Account, and we note that “lattice plasmon” (LP) is another common term for plasmonic SLRs. Theory predicts and recent experiments confirm that SLRs are delocalized over substantial distances (many tens to over $100 \mu\text{m}$).⁹ Furthermore, the resonance frequencies of SLRs are highly tunable from the near-IR to the UV depending on the size, shape, and material composition of the NPs, the lattice spacing and symmetry, and the refractive index environment of the surrounding me-

dium.^{7,8,10,11} Optimized SLRs exhibit quality factors (QFs) significantly higher (hundreds) than those of single-particle LSPs.

The unique combination of properties possessed by SLRs (high QFs, intensely localized electric fields, and micrometer-scale delocalization) has been harnessed to mediate nanoscale lasing phenomena.¹² In plasmonic lattices, a low-threshold lasing cycle is achieved by introducing a high-concentration liquid dye solution into the NP array to serve as the gain medium, while the SLRs play the role of the cavity modes in a traditional lasing system. The dye molecule is selected to ensure that its emission profile spectrally overlaps with an SLR. The combined dye–lattice system is driven with a femtosecond pump pulse at the absorption frequency of the dye, and a portion of the red-shifted light emitted by the dye subsequently couples to the SLRs. Dye molecules in close proximity to the NP surfaces experience the intense electric near-fields associated with LSPs, which leads to Purcell enhancement of the stimulated dye emission and hence lasing.

This Account is written for nanoscience theorists and experimentalists looking to learn the basic theoretical tools needed to understand and predict the optical properties of SLRs. We cover the theory at an introductory level by discussing the existence of SLRs in idealized 1D and 2D plasmonic lattices using the coupled dipole method. Also presented is an overview of semiquantum methods for modeling SLR lasing nanosystems. In addition, we include a survey of the tools needed to go further into the analysis of SLRs and provide an overview of past and current theoretical advances in the field.

■ COUPLED DIPOLE MODEL OF SLRS

Theoretical studies of periodic arrays of plasmonic NPs date back nearly four decades to the work of Meier, Wokaun, Liao, Schatz, and co-workers who investigated these systems for the purposes of surface-enhanced Raman scattering (SERS).^{13–15} Later, Markel¹⁶ explored the mathematical structure of infinite 1D chains of electromagnetically interacting dielectric scatterers using the coupled dipole theory of Purcell and Pennypacker.⁵ Markel was the first to obtain an analytic expression for the extinction cross section of an infinite chain of dielectric cylinders. In the context of LSP-supporting NPs, Schatz and co-workers applied the coupled dipole method to study collective excitations in 1D and 2D plasmonic arrays and were the first to report the possibility of achieving extinction features orders of magnitude narrower than those of an isolated NP LSP.^{7,8,17} Experimental confirmation of these effects was provided independently in 2008 by two groups studying 2D rectangular Au NP arrays.^{18,19}

1D Lattices

While important differences between 1D and 2D SLRs exist, the basic physical principles can be understood in the context of a 1D system. As shown in Figure 1a, to construct an SLR-supporting system, an array of metallic nanospheres of radius R may be arranged into a 1D lattice of periodicity a . At lowest order, the NPs interact via their induced LSP dipole moments $\mathbf{p}_n$. To model this interaction using the coupled dipole method, it is sufficient to first examine the dipole moment of a single NP in the array. In the time domain, the dipole moment of the n th particle is

$$\mathbf{p}_n(\mathbf{r}_n, t) = \int_{-\infty}^{\infty} dt' \tilde{\alpha}_n(t - t') \Theta(t - t') \mathbf{E}(\mathbf{r}_n, t') \quad (1)$$

where $\tilde{\alpha}_n(t - t')$ is the polarizability response function dyad of the NP and $\mathbf{E}(\mathbf{r}_n, t')$ is the total applied field at its center $\mathbf{r}_n$. The presence of the step function $\Theta(t - t')$ ensures that causality is obeyed. Use of the convolution theorem allows eq 1 to be expressed in frequency space as

$$\tilde{\mathbf{p}}_n(\mathbf{r}_n, \omega) = \tilde{\alpha}_n(\omega) \tilde{\mathbf{E}}(\mathbf{r}_n, \omega) \quad (2)$$

where frequency-space quantities are denoted as $\mathcal{F}[A(t)] \equiv \tilde{A}(\omega)$. The total applied electric field contains two types of terms. One originates from any external driving field, $\tilde{\mathbf{E}}_0(\mathbf{r}_n, \omega)$. The other is due to the induced electric fields from the other m dipoles in the array: $\tilde{\mathbf{E}}_{\text{scatt}}(\mathbf{r}_n) = (k_0^2/\epsilon_0\epsilon_r) \sum_m \tilde{\mathbf{G}}_{nm}(\mathbf{r}_n, \mathbf{r}_m, k_0) \tilde{\mathbf{p}}_m$, which is written in terms of the free-space dyadic Green's function

$$\tilde{\mathbf{G}}_{nm}(\mathbf{r}_n, \mathbf{r}_m, k_0) = \left[\left(1 + \frac{i}{k_0 R_{nm}} - \frac{1}{(k_0 R_{nm})^2} \right) \tilde{\mathbf{I}} - \left(1 + \frac{3i}{k_0 R_{nm}} - \frac{3}{(k_0 R_{nm})^2} \right) \hat{\mathbf{n}}_{nm} \hat{\mathbf{n}}_{nm} \right] \frac{e^{ik_0 R_{nm}}}{4\pi R_{nm}} \quad (3)$$

where $k_0 = 2\pi\sqrt{\epsilon_r}/\lambda = \sqrt{\epsilon_r}\omega/c$ is the magnitude of the wave vector in a medium of relative permittivity ϵ_r , $R_{nm} = |\mathbf{r}_n - \mathbf{r}_m|$ is the distance between the observation and source points $\mathbf{r}_n$ and $\mathbf{r}_m$, $\hat{\mathbf{n}}_{nm} = (\mathbf{r}_n - \mathbf{r}_m)/R_{nm}$ is the unit vector connecting them, and $\tilde{\mathbf{I}}$ is the identity matrix. For simplicity, we set $\epsilon_r = 1$.

Thus, the total applied field at the location of the n th NP is $\tilde{\mathbf{E}}(\mathbf{r}_n, \omega) = \tilde{\mathbf{E}}_0(\mathbf{r}_n, \omega) + \tilde{\mathbf{E}}_{\text{scatt}}(\mathbf{r}_n, \omega)$, yielding

$$\tilde{\mathbf{p}}_n(\mathbf{r}_n, \omega) = \tilde{\alpha}_n(\omega) \left[\tilde{\mathbf{E}}_0(\mathbf{r}_n, \omega) + (k_0^2/\epsilon_0\epsilon_r) \sum_m' \tilde{\mathbf{G}}_{nm}(\mathbf{r}_n, \mathbf{r}_m, \omega) \tilde{\mathbf{p}}_m(\mathbf{r}_m, \omega) \right] \quad (4)$$

where the prime on the summation denotes the exclusion of the $n = m$ term of the series. As the remainder of the discussion is carried out in the frequency domain, for simplicity, we drop the use of the tilde (i.e., $A(\omega) \equiv A(\omega)$).

With $\lambda = x, y, z$ and $g_{n-m}^\lambda(\omega) = (k_0^2/\epsilon_0\epsilon_r) \hat{\mathbf{e}}_\lambda \cdot \tilde{\mathbf{G}}_{nm}(\omega) \cdot \hat{\mathbf{e}}_\lambda$, the equation of motion for any single dipole orientation reduces to

$$p_n^\lambda(\omega) = \alpha(\omega) \left[E_0 e^{ikna} + \sum_m' g_{n-m}^\lambda(\omega) p_m^\lambda(\omega) \right] \quad (5)$$

where k is the wave vector representing propagation along the 1D chain. Moreover, as all three Cartesian dipole moments are uncoupled because of the dipole–dipole selection rules of 1D systems, only one dipole orientation need be considered. By defining the discrete forward and inverse Fourier transform pair $p_k^\lambda(\omega) = 1/\sqrt{N} \sum_n p_n^\lambda(\omega) e^{ikx}$, $p_n^\lambda(\omega) = 1/\sqrt{N} \sum_k p_k^\lambda(\omega) e^{-ikna}$ and ignoring the external driving field, eq 5 can be expanded into Bloch states to give

$$p_k^\lambda(\omega) = 2\alpha(\omega) \sum_{\Delta=1}^{\Delta_{\max}} g_{\Delta}^\lambda(\omega) \cos(\Delta ka) p_k^\lambda(\omega) \quad (6)$$

In this context, SLRs are defined as the self-sustaining normal modes of the NP array, with eigenfrequencies that satisfy the following condition:

$$\frac{1}{\alpha(\omega)} - \sum_{\Delta=1}^{\Delta_{\max}} 2g_{\Delta}^\lambda(\omega) \cos(\Delta ka) = 0 \quad (7)$$

The lattice sums $S_{ii} \equiv 2 \sum_{\Delta=1}^{\Delta_{\max}} g_{\Delta}^\lambda(\omega) \cos(\Delta ka)$ can be summed over an infinite number of nearest-neighbor interactions $\Delta_{\max} \rightarrow \infty$ to give analytic expressions for transversely (S_{xx}) and longitudinally (S_{yy}) oriented dipoles, yielding

$$S_{xx} = \frac{1}{4\pi a^3} \{ a^2 k_0^2 \ln[(1 - e^{ia(k_0-k)}) (1 - e^{ia(k_0+k)})] - iak_0 \text{Li}_2(e^{ia(k_0-k)}) - iak_0 \text{Li}_2(e^{ia(k_0+k)}) - \text{Li}_3(e^{ia(k_0-k)}) - \text{Li}_3(e^{ia(k_0+k)}) \} \quad (8)$$

and

$$S_{yy} = -\frac{1}{2\pi a^3} [iak_0 \text{Li}_2(e^{ia(k_0-k)}) + iak_0 \text{Li}_2(e^{ia(k_0+k)}) + \text{Li}_3(e^{ia(k_0-k)}) + \text{Li}_3(e^{ia(k_0+k)})] \quad (9)$$

where $\text{Li}_s(x) = \sum_n x^n/n^s$ defines a polylogarithm of order s . For longitudinal modes S_{yy} (blue traces in Figure 1b), the lattice sum quickly converges as a function of $\Delta_{\max}$, with little difference between $\Delta_{\max} = 1$ and $\Delta_{\max} = \infty$. This behavior occurs because of the longitudinal polarization of S_{yy} , which corresponds to head-to-tail configurations of dipoles, and thus, the dominant terms in the series are due to the near and intermediate terms of the dyadic Green's function presented in eq 3. Physically, as dipoles

radiate perpendicular to their axis of oscillation, far-field (and long-range) photons do not play a role in coupling between head-to-tail-oriented dipoles. This is in stark contrast to the transversely polarized lattice sum S_{xx} shown in Figure 1b for $k = 0$ and varying $\Delta_{\max}$ (red traces), where noticeable differences arise even at $\sim 10^2$ nearest-neighbor interactions.

The sharp peak that emerges at infinite-order interactions is in fact the lattice Bragg mode, and one way to interpret the existence condition of an SLR is to plot the real part of the inverse polarizability of the NP (α^{-1}) against the lattice sum (Figure 1c). Here the polarizability is taken from Mie theory, and the material is modeled using the Drude model²⁰ by setting the dielectric function of Ag to $\epsilon(\omega) = \epsilon_{\infty} - \omega_p^2/(\omega^2 + i\gamma\omega)$. Lattice modes are found by searching for intersections between the two curves. Importantly, because both absorptive and radiative losses occur, the system is intrinsically non-Hermitian, and eq 7 possesses both real and complex parts. The resulting eigenenergies are therefore necessarily complex-valued, and each SLR is a “quasi-mode” with a finite lifetime that is inversely proportional to the imaginary part of the eigenvalue. Thus, while it is often the case that only the real part of eq 7 is considered when searching for eigenfrequencies, in actuality both the real and imaginary parts of eq 7 (inset of Figure 1c) must equal zero. Moreover, despite the discrete nature of the system, a continuum of SLR modes of energy $E_{LP}(k) = \hbar\omega_{LP}(k)$ exists for every value of ka in the range $-\pi \leq ka \leq \pi$ in the limit of $N \rightarrow \infty$.

While calculation of the band structure exactly requires solving for complex eigenmodes, it may be indirectly accessed by calculating (or measuring experimentally) the extinction efficiency per NP of the array, defined as the extinction cross section for the 1D wave vector $\mathbf{k} = k\hat{\mathbf{y}}$ normalized to the product of the geometric cross section πR^2 and the square of the magnitude of the incoming electric field $|\mathbf{E}_0|^2$:

$$\bar{\sigma}_{\text{ext}}(\mathbf{k}) = \frac{k_0}{\epsilon_0 \pi R^2 |\mathbf{E}_0|^2} \text{Im}\{\mathbf{E}_0^* \cdot [\tilde{\alpha}^{-1} - \tilde{\mathbf{S}}]^{-1} \cdot \mathbf{E}_0\} \quad (10)$$

For transversely polarized SLRs ($S_{ii} = S_{xx}$), the extinction efficiency is shown in Figure 1d and is compared against $\text{Re}[\hbar\omega_{LP}(k)]$ for two cases: (1) eigenfrequencies derived from the real parts of eq 7 (blue dots) and (2) eigenfrequencies derived by also taking the imaginary part of eq 7 into account (white dots). While both methods lead to an SLR mode at $\text{Re}[\hbar\omega_{LP}(k=0)] \sim 2.62$ eV that is nearly degenerate with the Bragg mode, solving for the fully complex eigenfrequencies leads to a significantly different band structure. Furthermore, by the use of only $\text{Re}[\alpha^{-1} - S_{xx}] = 0$, SLR lifetimes cannot be calculated. When included for the mode at $k = 0$, $\text{Im}[\hbar\omega_{LP}(k=0)] \sim 3.5 \times 10^{-3}$ eV, in line with results produced using numerical methods.

2D Lattices

Before carrying out the coupled dipole analysis of a 2D lattice, we first examine the dispersion properties of lattice systems in the limiting case of vanishing NP size. In this limit, known as the empty-lattice or free-photon approximation,^{11,21} free photons are constrained only by the underlying symmetry of the lattice. To derive the empty-lattice dispersion relation, the wave equation for the magnetic field $\mathbf{H}(\mathbf{r}, \omega)$ in the frequency domain

$$\nabla \times \frac{1}{\epsilon} \nabla \times \mathbf{H}(\mathbf{r}) = \left(\frac{\omega}{c}\right)^2 \mathbf{H}(\mathbf{r}) \quad (11)$$

must be solved subject to periodic boundary conditions imposed by the symmetry of the lattice. Because of the translational symmetry within the xy plane, the magnetic field is expanded in terms of Bloch states $\mathbf{H}_k(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\mathbf{u}(\mathbf{r})$, where $\mathbf{k}$ is the Bloch vector and $\mathbf{u}$ is a real-space envelope function with the same periodicity as the lattice, i.e., $\mathbf{u}(\mathbf{r}) = \mathbf{u}(\mathbf{r} + \mathbf{R})$ for any lattice vector $\mathbf{R} = n\mathbf{a}_1 + m\mathbf{a}_2$. Substituting this expression into eq 11 and requiring that field components be transverse to the propagation direction [i.e., $(i\mathbf{k} + \nabla) \cdot \mathbf{u}_k(\mathbf{r}) = 0$] yields

$$k^2 \mathbf{u} - 2i\mathbf{k} \cdot \nabla \mathbf{u} - \nabla^2 \mathbf{u} = \epsilon \left(\frac{\omega}{c}\right)^2 \mathbf{u} \quad (12)$$

The periodicity of $\mathbf{u}$ allows it to be written in terms of an expansion over reciprocal lattice vectors $\mathbf{G} = n'\mathbf{b}_1 + m'\mathbf{b}_2$ to give $\mathbf{u}(\mathbf{r}) = \sum_{\mathbf{G}} \sum_{\lambda} g_{\mathbf{G}\lambda} e^{i\mathbf{G}\cdot\mathbf{r}} \hat{\mathbf{e}}_{\lambda}$, where $g_{\mathbf{G}\lambda}$ is the expansion coefficient associated with reciprocal lattice vector $\mathbf{G}$ and polarization direction λ . When this expression is plugged into eq 12 and the case of in-plane propagation for transverse magnetic (TM) ($\mathbf{H} \perp \hat{\mathbf{z}}$) and transverse electric (TE) ($\mathbf{H} \parallel \hat{\mathbf{z}}$) polarization is considered, the empty-lattice dispersion relation takes the form

$$|\mathbf{k}_{\parallel} + \mathbf{G}| = \sqrt{\epsilon} \left(\frac{\omega}{c}\right) \quad (13)$$

There are five types of Bravais lattices in two dimensions, each characterized by a distinct set of reciprocal lattice vectors $\mathbf{b}_1$ and $\mathbf{b}_2$ and a set of primitive lattice vectors $\mathbf{a}_1$ and $\mathbf{a}_2$ (Figure 2). The empty-lattice dispersion of each Bravais lattice is completely determined by its reciprocal lattice vectors.

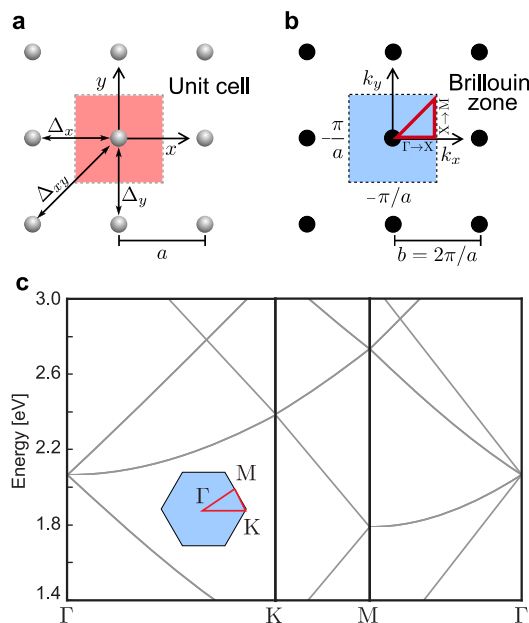


Figure 2. Real- and reciprocal-space representations of 2D NP arrays. (a) Schematic real-space representation of a square lattice with a unit cell shaded in red. Δ_{xy} , Δ_x , and Δ_y indicate the orders of nearest-neighbor interactions, though only $\Delta = 1$ is shown explicitly. (b) Reciprocal-space representation of the lattice shown in (a). The Brillouin zone is shaded in blue, and the irreducible Brillouin zone is outlined in red and connects the high-symmetry points Γ , X , and M . (c) Empty-lattice dispersion diagram for the diffractive orders supported by a hexagonal lattice with $|a_1| = |a_2| = 692$ nm.

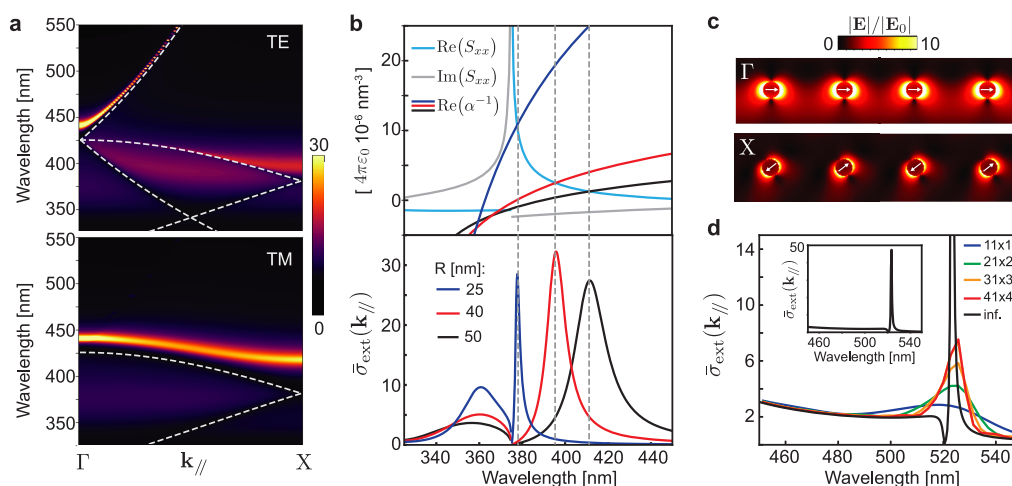


Figure 3. SLRs in 2D NP arrays. (a) Extinction efficiencies for an $a = 425$ nm cubic lattice of $R = 50$ nm Ag NPs excited by (top) TE- and (bottom) TM-polarized plane waves with incident directions corresponding to the Γ –X pathway through the Brillouin zone. The dielectric function of Ag was taken from experiment.²² Overlaid dashed lines show the empty-lattice dispersion. (b) Lattice sum and inverse polarizabilities (top) and extinction efficiencies (bottom) of the Γ -point SLR in an $a = 375$ nm square lattice as a function of NP radius R . (c) Cross sections of electric field enhancement in the xz plane for the lattice shown in (a) excited at the Γ and X points with TM-polarized plane waves. Field plots were calculated using the FDTD method. (d) Comparison of Γ -point SLR line shapes in the case of infinite and finite 2D square lattices ($a = 520$ nm; $R = 50$ nm Ag NPs).

The empty-lattice dispersion diagram can be compared to the band structure derived from the coupled dipole method, which yields an analogue to eq 7 of the form

$$\begin{pmatrix} \frac{1}{\alpha(\omega)} - S_{xx}(\omega) & -S_{xy}(\omega) & -S_{xz}(\omega) \\ -S_{yx}(\omega) & \frac{1}{\alpha(\omega)} - S_{yy}(\omega) & -S_{yz}(\omega) \\ -S_{zx}(\omega) & -S_{zy}(\omega) & \frac{1}{\alpha(\omega)} - S_{zz}(\omega) \end{pmatrix} \begin{pmatrix} p_{k_{\parallel}}^x \\ p_{k_{\parallel}}^y \\ p_{k_{\parallel}}^z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (14)$$

Here the 2D lattice sums for a square lattice (Figure 2a) reduce to

$$S_{ij} = \sum_{\Delta_{xy}=1}^N 4g_{\Delta_{xy}}^{ij} \cos(\Delta_{xy}k_x a) \cos(\Delta_{xy}k_y a) + \sum_{\Delta_x=1}^N 2g_{\Delta_x}^{ij} \cos(\Delta_x k_x a) + \sum_{\Delta_y=1}^N 2g_{\Delta_y}^{ij} \cos(\Delta_y k_y a) \quad (15)$$

where $i, j = x, y, z$, $g_{\Delta_{\lambda}}^{ij}(\omega) = \hat{\mathbf{e}}_i \cdot (k_0^2 / \epsilon_0 \epsilon_r) \vec{\mathbf{G}}_{nm}(\omega) \cdot \hat{\mathbf{e}}_j$, and $\lambda = xy, x, y$ (Figure 2a). In this form, the second and third terms emerge from the coupling among NPs along the x direction (Δ_x), and the y direction (Δ_y), respectively, while the first term emerges from coupling between diagonal nearest-neighbor elements (Δ_{xy}). There are several important differences between 1D and 2D SLR systems. From a physical point of view, all three Cartesian components of $\mathbf{p}_n$ are decoupled in one dimension, while only the out-of-plane z component is decoupled in two dimensions. Consequently, normal modes in 2D lattices cannot simultaneously have both in- and out-of-plane dipole moment components for any lattice symmetry. Mathematically, the lattice sums in the 2D case are only conditionally convergent, and closed-form representations are not known to exist. Brute-force evaluation of lattice sums is inefficient because of the long-

range $1/R$ dependence of the dyadic Green's function (eq 3), though mathematical techniques have been developed to accelerate their convergence. One popular method, originally developed by Ewald, involves splitting the lattice sum into two exponentially converging sums, one in the spatial domain and the other in the spectral domain.^{23,24} While it can be argued that all physically realizable systems are finite, state-of-the-art fabrication methods are capable of creating lattices with $\sim 10^2$ nm scale periodicity over square centimeter areas with $\sim 10^{12}$ (trillion) NPs, making such systems effectively infinite.

As in the 1D case, the band structure of a 2D lattice can be accessed indirectly by forcing the system with plane wave light and mapping out $\bar{\sigma}_{\text{ext}}(\mathbf{k}_{\parallel})$ as a function of $\mathbf{k}_{\parallel}$. Figure 3a shows the $\mathbf{k}_{\parallel}$ -dependent extinction efficiencies for TE- and TM-polarized plane waves along the $\Gamma(\mathbf{k}_{\parallel} = 0\hat{\mathbf{x}} + 0\hat{\mathbf{y}}) \rightarrow \text{X}(\mathbf{k}_{\parallel} = \frac{\pi}{a}\hat{\mathbf{x}} + 0\hat{\mathbf{y}})$ path through the Brillouin zone, with the empty-lattice dispersion from eq 13 overlaid with dashed white lines. The coupled dipole lattice dispersion largely follows that predicted by the empty-lattice approximation except near high-symmetry points, where Bragg scattering leads to the formation of band-edge states analogous to what is observed in electronic systems.²¹

Deviations from the empty-lattice dispersion that arise from scattering off LSPs at each lattice site are encoded into the form of the NP polarizability $\alpha(\omega)$, which we take from Mie theory.²⁰

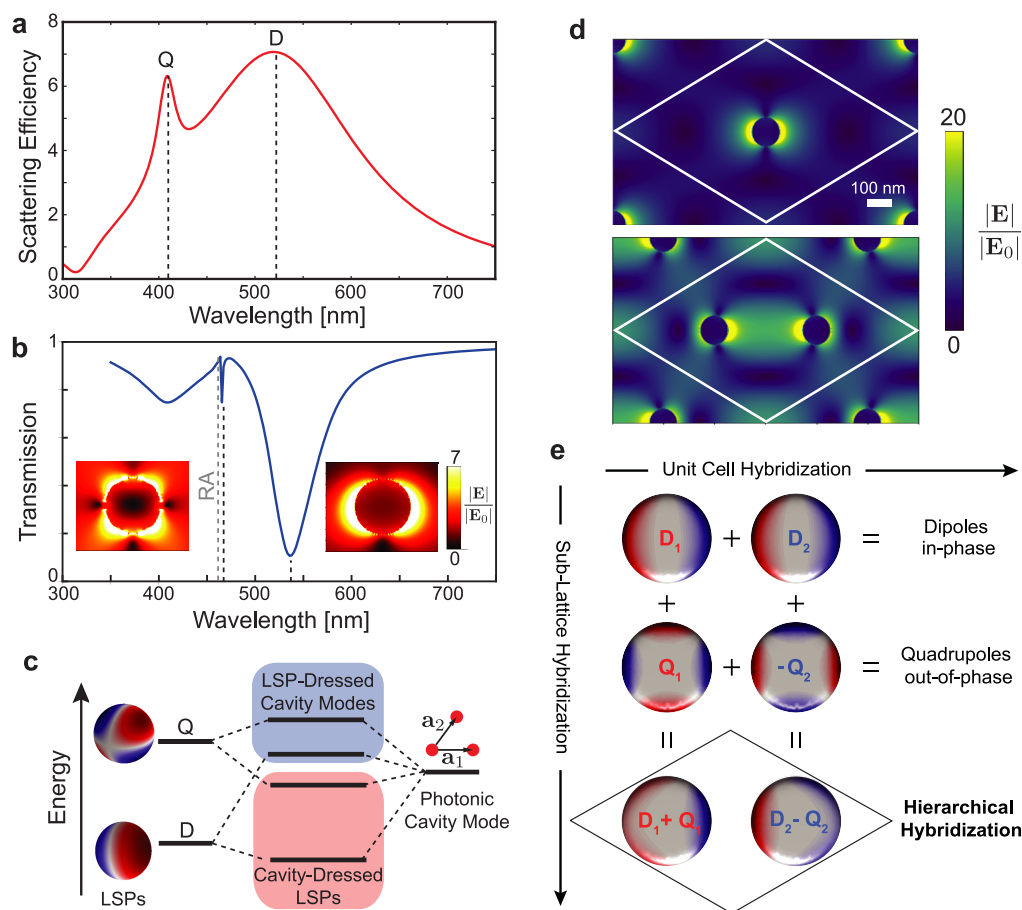


Figure 4. Quadrupole SLRs. (a) Ag NP scattering efficiency spectrum showing electric dipole (D) and quadrupole (Q) LSP modes. (b) Γ -point SLR transmission spectrum showing narrow-linewidth quadrupole SLR calculated using FDTD ($a = 320$ nm, $R = 50$ nm Ag NPs, $n = 1.45$). The insets show near-field profiles of electric fields at wavelengths corresponding to dipole and quadrupole SLRs. (c) Energy diagram illustrating the hybridization between dipole and quadrupole LSPs with photonic cavity modes. (d) Field enhancement maps associated with excitation of Γ -point SLRs in arrays with (top) hexagonal and (bottom) honeycomb geometries. While the field exhibits a dipolar distribution for the hexagonal lattice, the asymmetry shown in the fields surrounding the NPs in the non-Bravais honeycomb lattice is indicative of contributions from higher-order multipolar LSPs. (e) Hierarchical hybridization scheme showing LSP hybridization at both the NP and unit-cell levels. Panels (d) and (e) adapted from ref 29. Copyright 2019 American Chemical Society.

As in the 1D case, SLR eigenenergies can be approximated by solving $\text{Re}[\tilde{\alpha}^{-1} - \tilde{S}] = 0$. Because the scattering cross section scales as R^6 and it is often the case that the energy associated with the photonic mode is selected to be lower than that of the NP LSP, the Γ -point SLR red-shifts as the NP radius increases (Figure 3b). Inspection of the top panel of Figure 3b shows that there are two such intersection points, one above and another below the Rayleigh anomaly (RA), where the lattice sum diverges. The wavelengths associated with these intersections correspond to the resonant energies of the hybridized LSP cavity modes. For the case shown in Figure 3b, the cavity-dressed LSPs appear near 360 nm, and the LSP-dressed cavity modes (i.e., SLRs) always appear to the red of the RA (above 375 nm). The composition of the resulting SLR modes can therefore be tuned to possess either a more plasmonic or photonic behavior by varying the NP size. This effect is most evident in the resulting linewidths and the corresponding lifetimes of the modes in question. It should be noted that spherical NPs have been used here for simplicity, while the use of anisotropic NPs (e.g., cylinders) leads to additional lifting of the band degeneracies at the high-symmetry points of the Brillouin zone due to the anisotropic polarizability of the NP.

To test the robustness of the coupled dipole method, full-wave solutions to Maxwell's equations on an infinite lattice were calculated using the finite-difference time domain (FDTD) method, as shown in Figure 3d. The parameters of the lattice and the NP were chosen such that the system is simultaneously optimized for high-QF Γ - and X-point excitations. Here we see in Figure 3c that, as predicted from the analysis presented above, the Γ -point SLR is red-shifted from the RA at 425 nm, occurring instead at 440 nm, and all of the dipole moments are oriented in-phase along the direction of the incident field polarization ($|\mathbf{k}_{\parallel}| = 0$). For excitation at the X point, again consistent with the predictions of the coupled dipole method, the SLR resonant frequency occurs at 412 nm, and adjacent dipoles are exactly π out of phase with each other in the direction of $|\mathbf{k}_{\parallel}| = \pi/a$. Even though the resulting SLR is nearly degenerate with the RA, the hybridized nature of the system allows for the existence of a mode with the characteristic high near-field enhancement of LSPs in addition to the narrow line shape expected from a photonic crystal mode.

The system of eqs 4 can also be solved using real-space Wannier states (site-space) instead of Bloch states for an array of area $(Na)^2$. In this case, difficulties arising from efficient evaluation of the lattice sum are avoided at the expense of

following $3N$ degrees of freedom. Inversion of a $3N \times 3N$ matrix is then required to solve for the orientation of each NP's dipole moment. Figure 3d shows how the extinction efficiency spectra of an $a = 520$ nm square lattice evolves with N . As the number of NPs in the lattice increases, the SLR linewidth decreases, ultimately converging to that of the infinite lattice at large N .²⁵ Although the spectral evolution of the finite lattice system as N increases is reminiscent of what was observed in the 1D case, the two cases are physically distinct. An infinite lattice was assumed in the discussion of nearest neighbors in the 1D chain, whereas the evolution considered here arises in the case of a finite lattice where each NP is coupled to every other NP in the lattice. The above presentation of the coupled dipole method is valid for any 2D Bravais lattice with a single NP per unit cell and all NPs identical, but it has been extended to accommodate more complicated lattice geometries with more than one NP per unit cell.^{26–28}

Beyond the Dipole Approximation

While the coupled dipole method provides a simple physical picture and analytic framework for understanding collective excitations in plasmonic lattices, it is not capable of describing contributions from higher-order LSPs. Related coupled multipole methods have been developed that account for these additional degrees of freedom (magnetic dipole and electric quadrupole, for example). In situations where the coupled dipole method is insufficient, full-wave electrodynamics methods can be employed. Many such computational electrodynamics methods have been developed, such as the frequency-domain T-matrix method, but they are applicable only to systems with a finite number of particles. The boundary element method (BEM)³⁰ and the FDTD method³¹ can also be used.

As shown in Figure 4, for lattices composed of NPs with dimensions D such that $2\pi D/\lambda \ll 1$ is not satisfied, higher-order multipolar LSPs can be sustained. Indeed, spherical multipoles form the basis of Mie theory, and for relatively large NPs or distances near their surface, multipolar LSPs can even dominate the dynamics. To include these effects, multipolar decomposition techniques exist to expand the electromagnetic fields scattered by nonspherical objects in the spherical multipole basis. The coupled dipole method has also been extended to include dipole–quadrupole and quadrupole–quadrupole interactions to predict the existence of SLRs mediated by quadrupolar LSPs in 1D and 2D lattices.^{32,33} Quadrupole SLRs have been observed experimentally^{10,34} in linear transmission spectra with Γ -point energies and line shapes well-described by the coupled dipole–quadrupole method.³³ The FDTD and T-matrix methods have also been used to study the excitation of “dark” quadrupole SLR modes excited by using a finite lattice to relax the infinite-lattice selection rules.³⁵ Moreover, SLR-supporting non-Bravais lattices have been shown to exhibit novel mode-mixing mechanisms such as hierarchical hybridization, which accounts for the unique optical properties of honeycomb lattices. The formation of SLRs mediated by multiorder LSPs can result in asymmetric electric near-field distributions surrounding the NPs (Figure 4d,e). This asymmetry is due to LSP hybridization at the individual NP level from LSPs of different orders and at the unit cell level (NP dimer) from LSPs of the same order.²⁹

THEORY OF SLR LASING

Miniaturized lasers employing plasmonic lattices are an emerging platform for generating coherent light for on-chip

photonic circuits, large-scale quantum photonics, solid-state lighting, and in vivo cellular imaging.^{36–38} Beyond the diffraction limit, which currently constrains the effective mode volumes of dielectric cavities, subwavelength plasmon nanolasing can be achieved by exploiting band-edge SLRs in metal NP arrays as the optical feedback mechanism of a lasing system, with dye molecules playing the role of the gain medium.^{12,34,39} A commonly used model for SLR lasers is shown in Figure 5,

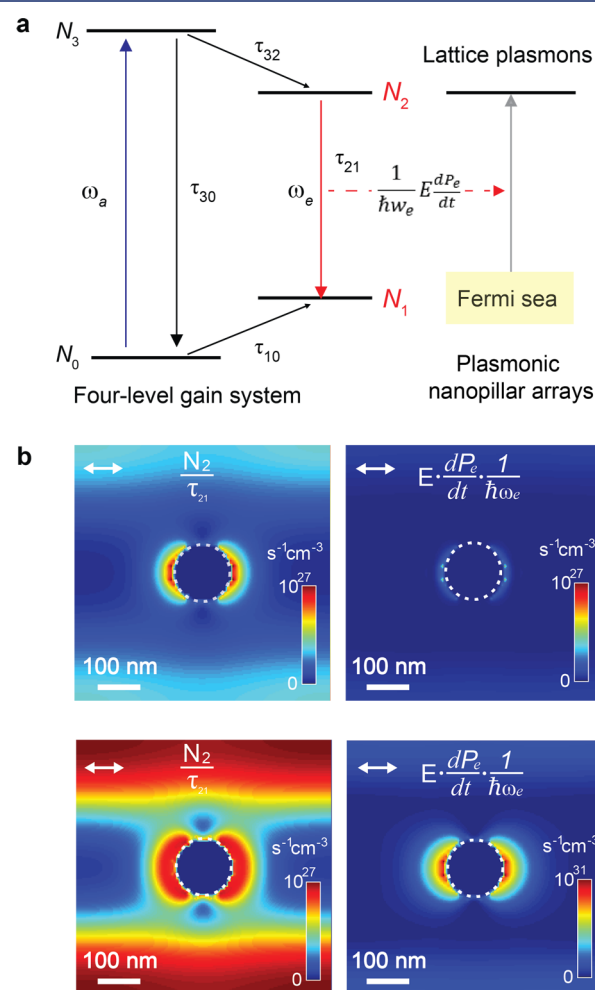


Figure 5. Modeling of nanoscale energy transfer with a semiquantum four-level system for lasing action. (a) Scheme of the energy transfer process from the four-level gain medium to SLR resonances in sparse NP arrays. (b) Calculated distribution maps of the (left) spontaneous and (right) stimulated transition rates in the middle plane of NP arrays (top) below and (bottom) above threshold. The unit is $s^{-1} cm^{-3}$. Figure adapted from ref 12. Copyright 2013 Nature Publishing Group.

where the dye solution is approximated as a four-level system, and emission and absorption associated with the dye molecules is coupled to the lattice using electrodynamics.¹² The explicit time dynamics for the population densities of all four levels, the transitions between levels, and the electromagnetic response of the plasmonic lattice is then obtained self-consistently. One approach^{40–42} is to use a semiquantum model that treats the transition dipole moments between the optically active transitions as a set of coupled oscillators. In this system, we assume that the transition dipoles are induced by the electromagnetic field on the basis of a set of driven equations of motion as follows:

$$\begin{aligned}\frac{d^2\mathbf{P}_e}{dt^2} + \gamma_e \frac{d\mathbf{P}_e}{dt} + \omega_e^2 \mathbf{P}_e &= \xi_e(N_2 - N_1)\mathbf{E}(t) \\ \frac{d^2\mathbf{P}_a}{dt^2} + \gamma_a \frac{d\mathbf{P}_a}{dt} + \omega_a^2 \mathbf{P}_a &= \xi_a(N_3 - N_0)\mathbf{E}(t).\end{aligned}\quad (16)$$

where $\mathbf{E}$ is the electric field due to the plasmonic NP lattices and the external driving field and $\mathbf{P}_i$ is the polarization field resulting from the i th optically active transition. The transitions between population densities N_0 (N_2) and N_3 (N_1) are parametrized by an empirically derived transition frequency ω_e (ω_a), damping constant γ_e (γ_a), and coupling efficiency $\xi_i = 6\pi\epsilon_0 c^3/\omega_i \gamma_i$. Transitions taking the system from N_3 to N_2 and N_0 to N_1 are not optically active and are thus not explicitly considered in this expression. However, transitions leading to changes in the populations of all of the states are described using a standard rate equation approach as follows:

$$\begin{aligned}\frac{dN_3}{dt} &= -\frac{N_3(1-N_2)}{\tau_{32}} - \frac{N_3(1-N_0)}{\tau_{30}} + \frac{1}{\hbar\omega_a}\mathbf{E}(t) \cdot \frac{d\mathbf{P}_a}{dt} \\ \frac{dN_2}{dt} &= -\frac{N_3(1-N_2)}{\tau_{32}} - \frac{N_2(1-N_1)}{\tau_{21}} + \frac{1}{\hbar\omega_e}\mathbf{E}(t) \cdot \frac{d\mathbf{P}_e}{dt} \\ \frac{dN_1}{dt} &= -\frac{N_2(1-N_1)}{\tau_{21}} - \frac{N_1(1-N_0)}{\tau_{10}} - \frac{1}{\hbar\omega_e}\mathbf{E}(t) \cdot \frac{d\mathbf{P}_e}{dt} \\ \frac{dN_0}{dt} &= -\frac{N_3(1-N_0)}{\tau_{30}} - \frac{N_1(1-N_0)}{\tau_{10}} - \frac{1}{\hbar\omega_a}\mathbf{E}(t) \cdot \frac{d\mathbf{P}_a}{dt}\end{aligned}\quad (17)$$

To close the system of equations, it is also necessary to include the interactions between the polarization fields and Maxwell's equations,

$$\begin{aligned}\nabla \times \mathbf{E} &= -\mu_0 \frac{\partial \mathbf{H}}{\partial t} \\ \nabla \times \mathbf{H} &= \epsilon_r \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \frac{\partial \mathbf{P}}{\partial t}\end{aligned}\quad (18)$$

which may be solved numerically using an extension of the FDTD method. To simulate lasing from the four-level system shown in Figure 5, the entire system is pumped from the ground state at a frequency ω_a with initial conditions $N_0 = 1$ and $N_1 = N_2 = N_3 = 0$, allowing the above equations along with Maxwell's equations to be solved in the time domain.

Using a variant of the semiquantum model above, the Schatz group has modeled lasing action in plasmonic NP arrays integrated with different gain media, and the results show very good agreement with the experiments.^{12,34,43} The modeling results confirm that lasing occurs at the same wavelength as the SLR resonance and can be directly modulated by the refractive index environment around the metal NPs.^{34,33} Moreover, the spatial- and time-dependent properties associated with lasing buildup can be calculated, predicting a spontaneous emission rate (N_2/τ_{21}) and stimulated emission rate ($\frac{1}{\hbar\omega}\mathbf{E}(t) \cdot \frac{d\mathbf{P}}{dt}$) that are spatially correlated with plasmonic hot spots around the NP surfaces. With the model described above, it was found that stimulated emission was primarily localized to dye molecules near the LSP dipole or quadrupole hot spots.^{12,34} Notably, dye molecules in the subwavelength vicinity of NPs (<50 nm) contributed most to the population inversion and hence the lasing emission.^{12,34} Inclusion of the Purcell effect can be

achieved by modeling the enhanced spontaneous emission rates of dipole emitters in the presence of plasmon nanocavities.⁴¹ In addition, the ultrafast dynamics of plasmon lasing was found to be sensitive to the cavity mode quality and pump power.^{39,44} With increased pump intensity, the model predicted an earlier rise time and a shorter decay time, consistent with experiments in single-lattice NP arrays.^{39,44}

To maximize computational resources and gain access to the suite of features available in commercially available electro-dynamics solvers, the model of the one-electron, four-level system described above was integrated with the FDTD-method-based software package Lumerical.³⁹ With Lumerical routines, it was possible to model off-normal emission in the plasmonic superlattices involving finite arrays of NPs (patches) grouped into microscale arrays.^{39,45} Also, simulations predicted and experiments confirmed that pumping the electronic ground state resulted in lasing modes from the band-edge Γ -point mode at $|\mathbf{k}_{||}| = 0$. Experiments also confirmed the theoretical predictions that multimodal lasing arises from excitation of a superlattice system with a plane wave source at oblique angles, leading to band-edge SLRs with nonzero wave vectors. With the semiquantum model integrated directly in Lumerical, the ability to characterize lasing action from different-sized lattices with complicated global symmetries, unit cell designs, NP morphologies, and arbitrary emission angles paves the way for studying light-matter interactions in a variety of plasmonic and photonic nanocavities.^{38,39,44} Furthermore, three types of lasing mechanisms (localized surface plasmons, pure photonic modes, and mixed photonic/plasmonic modes) were identified by tuning the spectral position of the Bragg mode relative to the LSP frequencies of a single NP.⁴¹ For a range of considered lattice constants, it was predicted that two lasing peaks with comparable thresholds and emission intensities could be simultaneously supported by using pure photonic and hybrid modes.

In an extension to the plasmon lattice nanolasing model presented above, the rate equations were replaced with quantum Liouville equations to determine the importance of quantum effects such as dephasing and non-Markovian dynamics on the lasing process.⁴⁶ Integration of the Maxwell–Liouville equations with FDTD calculations revealed the effect of dephasing in the quantum dynamics on the emission intensity and linewidth.⁴⁶ Including both electronic and vibrational dephasing pathways was found to reduce lasing emission but to have little effect on long-time population inversion. The simulation results using the Maxwell–Liouville method also exhibited a 2-fold enhancement in emission intensities and narrower linewidths compared with those predicted by the rate equation model.⁴⁰

RECENT THEORETICAL AND COMPUTATIONAL RESULTS

When examining the linear optical properties of SLRs, the role of quadrupole-mediated SLRs has been of particular interest.^{10,32–34} SLRs in non-Bravais lattice systems are also beginning to be investigated,^{26–28} although the study of multipole-mediated SLRs in these systems remains largely unexplored. Further, while many studies have focused only on Γ -point modes, designing lattices to explore other high-symmetry points will almost certainly be the focus of future research.

Recently, effort has been devoted to searching for the existence of topologically protected edge modes at the interface between plasmonic lattices of different symmetry or between the edge of a lattice and a continuous medium. While topological

states have been shown to exist in dielectric lattices, whether they exist in plasmonic lattices remains an open question because of complications arising from the non-Hermitian (lossy) nature of plasmonic systems. Several recent theoretical works investigated possible topological effects in 1D and 2D chains of NPs but considered only inter-NP distances that precluded the formation of SLRs.^{47–50} Recent interest in SLR band structure (spurred by interest in achieving topological effects) has also led to the observation of lasing from the K points of a plasmonic honeycomb lattice.⁵¹

One of the most enticing features of SLRs is their ability to combine the long lifetimes and propagation lengths associated with photonic cavity modes with the extraordinary near-field enhancement properties of LSPs. These features conspire to make plasmonic lattices an attractive platform for modifying the emission dynamics of nearby quantum emitters via the Purcell effect.^{52,53} Strong coupling between plasmonic SLRs and ensembles of quantum emitters has also been demonstrated, showing the characteristic frequency splitting in the emission spectra of both SLRs and quantum emitters.^{54–56} The bosonic nature and low effective masses of SLRs and SLR polaritons also led to the prediction of room-temperature Bose–Einstein condensation of SLRs interacting with a bath of quantum emitters,^{57,58} for which experimental verification was recently achieved.⁵⁹

CONCLUSIONS AND PERSPECTIVES

In this Account, we have overviewed the theoretical and computational aspects of plasmonic SLRs, beginning with a derivation of the basic properties of SLRs based on the coupled dipole method. This method allowed us to solve for the complex-valued SLR eigenfrequencies of a 1D chain of NPs, providing a connection between the SLR and Bragg mode energies and predicting the lifetime of the Γ -point SLR to be ~ 500 fs. For 2D lattices, we showed how the addition of another dimension complicates the propagation and optical properties of SLRs, giving rise to a much richer band structure. The empty-lattice model provides a simple scheme to understand SLR dispersion properties, but coupled dipole and even full-wave calculations are essential for quantitative analysis. Also, quadrupole SLR excitation has emerged as an important feature of plasmonic lattices that requires going beyond the coupled dipole approximation, which leads to narrow modes and richer dispersion properties than with purely dipolar SLRs. Finally, we outlined a semiquantum model of plasmonic lasing that provides an approach for coupling the electrodynamics of SLRs to laser dye photophysics. The model leads to an accurate description of many different types of plasmonic lasers that involve variation in lattice structure, nanoparticle size and shape, and gain material properties. Indeed, the semiquantum approach is now used as a tool for predicting the feasibility of new laser designs, thereby enhancing the effectiveness of new developments in the field.

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ACKNOWLEDGMENTS

We thank Teri Odom and her group for many years of collaboration that contributed to papers cited in this Account. The nonlaser theory part of this research was supported by DOE Grant DOE DE-SC0004752, while the laser theory work was supported by NSF Grant DMR-1608258.

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